

Education Tracks and Attitudes Towards Women's Participation in the Labour Force and Leadership

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Abstract

We estimate the causal effect of general versus vocational secondary education on gender role attitudes. Using a fuzzy regression discontinuity design that exploits a preparatory exam score cutoff, we find that general secondary education substantially increases support for gender equality. General secondary graduates are 16 percentage points more likely to agree that women should be allowed to work and 24 percentage points more likely to support women in leadership, relative to vocational graduates. These effects are larger among male graduates than female graduates. Exploring mechanisms, we find evidence that track assignment operates through differences in access to higher education, exposure to female coworkers, peer composition, and curriculum content. Our findings suggest that expanding social and civic content in vocational curricula and broadening higher education pathways for vocational graduates could promote more egalitarian gender attitudes and advance gender equality.

Keywords: Gender role attitudes, female labour force participation, female leadership, Vocational education

JEL Codes: A21; I20; J16

1 Introduction

Globally, far fewer women participate in the labour force than men. In 2024, 73% of men were in the labour force compared to only 49% of women¹. Among women who do participate in the labour market, an even smaller share reach leadership positions. In 2022, women held only 28 per cent of management positions globally (ILO, 2024) and comprised just 27 per cent of the world’s parliament members². One of the important determinants of gender gap in the labour markets and leadership positions is attitude towards gender roles³. Gender role attitudes⁴ (GRAs) play a critical role in perpetuating or mitigating these disparities by shaping individual biases and consequently decisions, organisational policies, and social norms (Lyness and Heilman, 2006; Eagly, 2007; Heilman et al., 2024). Understanding what determines these attitudes, particularly towards women’s workforce participation and leadership roles, is therefore essential for addressing persistent gender gap in labour markets.

Existing literature identifies various factors that influence the GRAs, including historical determinants such as religious doctrine (Seguino, 2011), political-economic regimes (Campa and Serafinelli, 2019), and the agricultural practices of the early settled communities (Alesina et al., 2013). Socio-demographic characteristics, including gender, ethnicity, immigration status, socioeconomic status, and age, further shape these attitudes (Kågesten et al., 2016; Patel et al., 2021; Yüçetas and Carol, 2024). These attitudes are also transmitted across generations through direct parent-child interactions and observational learning (Cunningham, 2001; Dhar et al., 2019). Subsequently, these inherited attitudes are reinforced or challenged through key social institutions, including workplaces, religious

¹These statistics can be accessed from the World Development Indicators (WDI): [here](#) and [here](#)

²This data can be accessed from the World Bank database: [here](#)

³Existing literature has discussed several demand and supply side factors for low female labour participation (see Jayachandran (2020) for a review). Factors that can explain gender gap in leadership include gender differences in competitiveness and risk aversion (Niederle and Vesterlund, 2011), the confidence gap between men and women (Kagel and Roth, 2020), social environment, institutional barriers, and discrimination (see Marianne (2011); Blau and Kahn (2017); Neumark (2018) for a review of literature).

⁴Gender role attitudes are an individual’s personal beliefs about what roles are appropriate for men and women, while gender norms are society’s shared, collectively enforced expectations about how men and women should behave (McHugh and Frieze, 1997).

organisations, media outlets, and educational systems ([Webb and Temple, 2016](#); [Meuleman et al., 2017](#); [Patel et al., 2021](#); [Dhar et al., 2022](#)).

In this paper, we causally estimate the effect of educational tracks during secondary school in shaping long term GRAs. Specifically, we estimate the effect of completing general rather than vocational secondary education on support for women’s participation in workforce and leadership roles. Attitudes towards both of these directly affect labour-market and organisational inequality. Egypt provides a compelling context for this analysis, as gender role attitudes in the country remain among the most traditional around the world According to the World Values Survey (2017 to 2022), 87% of Egyptians agree that men should have priority over women when jobs are scarce, 82% believe men make better political leaders, and 74% believe men make better executives.⁵ Further, focusing on educational tracks during secondary school is particularly salient because adolescence represents a critical period for value formation. During this developmental stage, individuals possess the cognitive capacity to engage with complex social and moral issues while also remaining more open to attitudinal change than in adulthood ([Kohlberg, 1976](#)). Therefore, understanding how different educational environments shape gender attitudes during this formative period has important implications for policy interventions aimed at promoting gender equality.

General and vocational tracks differ substantially in curriculum content, peer composition, and access to higher education ([Elbadawy, 2015](#)). These differences can shape how students develop attitudes towards gender roles. General secondary track provides broader academic training emphasising critical thinking and social inquiry, creating opportunities to engage with and question traditional norms. Vocational track, by contrast, prioritises technical skill development and offers limited exposure to discussions of social structures and gender equality. Moreover, general secondary graduates have substantially greater access to higher education ([Krafft, 2018](#)), which further exposes them to diverse perspectives and egalitarian values. Consequently, secondary educational tracks may produce divergent attitudes towards women’s labour market participation and leadership capabilities.

⁵These statistics can be accessed from World Value Survey’s online data analysis website: [Click here](#)

However, estimating the causal effect of the education track on these attitude outcomes is challenging due to endogeneity problem arising from self-selection into either general or vocational education track. To address this challenge, we use a fuzzy regression discontinuity design (FRDD) based on an exogenous eligibility cut-off for general secondary education track. In Egypt, students must score at least 70 points on their final preparatory school (9th grade) exam to qualify for the general secondary track. However, students with a score of 70 or above can choose to opt for vocational track and students with score below 70 can take general track in private schools⁶. The identification strategy relies on the fact that the cutoff of 70 makes the probability of completing general secondary education a discontinuous function of the preparatory exam score. Thereby, comparing students just below and above the cut-off can isolate the causal effect of secondary education track. For this purpose, we use data from the 2012 (OAMDI, 2020) and 2018 (OAMDI, 2019) rounds of Egypt labour market panel survey (ELMPS). We measure *gender role attitudes* along two dimensions through individual agreement with the statements: (i) “*A woman’s place is not only in the household, but she should also be allowed to work*”, (ii) “*women should continue to obtain leadership positions in society*”.

Using FRDD, we find that among secondary school graduates, general education graduates are about 16 percentage points more likely to agree that women should be allowed to work and roughly 24 percentage points more likely to support women holding leadership positions. The effect is higher among male graduates. The point estimates remain stable across various bandwidths of the running variable. They are also robust to the inclusion of other predetermined covariates that may influence education and attitudes towards women’s leadership roles. These covariates do not show significant differences at the cutoff. We also explore the mechanisms through which general and vocational secondary education may impact attitudes towards women’s leadership roles. Our analysis reveals that general secondary graduates are more likely to pursue post-secondary education, which may contribute to their more egalitarian views on women’s roles in work and leadership

⁶Private schools are costly and only about 6% of the general track students enrol in private schools (Krafft et al., 2019)

([Shu and Zhu, 2012](#)). In addition, general secondary graduates are more likely to work in environments with female colleagues, a factor associated with fostering more egalitarian gender role attitudes ([Meuleman et al., 2017](#); [Yadav and Saharan, 2025](#)).

Our paper contributes to multiple strands of literature. First, we extend research documenting the effects of education on gender role attitudes ([Dhar et al., 2022](#); [Yüçetas and Carol, 2024](#)). More specifically, we contribute to a growing body of work showing that school context, such as school type, classroom gender composition, and peer characteristics, plays a critical role in shaping gender role attitudes ([Erarslan and Rankin, 2013](#); [Kågesten et al., 2016](#); [Sánchez Guerrero and Schober, 2021](#); [Taraszow et al., 2024](#)). Unlike most existing research focusing on developed countries, we analyse a developing country context where gender norms are more traditional and educational systems differ substantially. Additionally, we provide causal estimates by employing a fuzzy regression discontinuity design that addresses self selection into educational tracks. Finally, we examine long-term attitudinal effects rather than immediate outcomes, revealing how educational tracking shapes beliefs that persist into adulthood.

Second, our study contributes to the literature on the effects of vocational education by moving beyond labour market outcomes to examine its broader societal implications ([Bol et al., 2019](#); [Brunetti and Corsini, 2019](#); [Hanushek et al., 2017](#); [Kemper and Renold, 2024](#)). Recent research in Egypt has examined how general versus vocational tracks affect labour market outcomes, finding that general education graduates experience advantages in formal employment ([Kemper and Renold, 2024](#)). We extend this line of inquiry by investigating how educational tracks shape attitudes towards women’s labour force participation and leadership. By doing so, we illuminate an unexplored mechanism through which vocational education may perpetuate gender disparities in labour markets and organisational hierarchies, complementing existing evidence on economic outcomes with insights into attitudinal formation.

The rest of the paper is structured as follows: Section 2 provides an overview of Egypt’s education system, followed by contextual background on attitudes towards women’s work

and leadership. Section 3 details the data and sample used in the analysis, while Section 3.2 outlines the empirical strategy. Section 4 presents the empirical findings, and Section 5 concludes the study.

2 Background

2.1 The Egyptian education system

The Egyptian school system consists of three main levels, namely primary, preparatory, and secondary education (see Figure 1). Compulsory education in Egypt begins at age six and lasts for nine years, encompassing six years of primary education (grades 1 to 6) followed by three years of preparatory education (grades 7 to 9) (Krafft et al., 2019). At the end of preparatory education, students must take a centralised national examination in which they must score at least 50 out of 100 points to pass the test (Kemper and Renold, 2024). The result of preparatory school exam determines whether students go to general or vocational secondary schools.

Students who score less than 70% on the preparatory school exam are typically placed on the vocational track, which is almost always a terminal degree, often leading directly to entry into the labour market (Krafft, 2018). Conversely, those who achieve scores of 70 or higher are admitted to the general secondary track, which facilitates a nearly universal progression to higher education, serving as a gateway to universities and other tertiary institutions (Assaad, 2010; Assaad and Krafft, 2015). However, the cut-off of 70% is not binding. Students with a score less than 70% can still attend the general education track at private secondary schools, although it is very costly and therefore not very common (about 6% of the general track students enrol in private schools (Krafft et al., 2019)). Similarly, those who get more than or equal to 70% can choose to go to vocational secondary school (Assaad, 2010). In Egypt, approximately one-third of secondary school graduates complete the general secondary track, while the remaining two-thirds graduate from the vocational

track (Assaad, 2010).

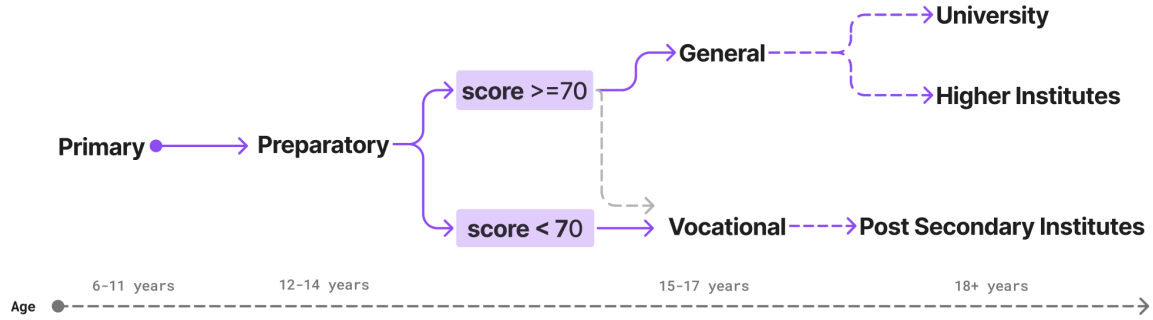


Figure 1: Education structure in Egypt

Note: Ages in parentheses indicate the ideal age, assuming on-time entry without repetition. Graphic based on Krafft et al. (2019). Post-secondary institutes typically offer two-year diplomas, whereas higher institutes and universities offer four-year degrees.

Students who successfully complete the three-year general secondary track are awarded the General Secondary Education Certificate (Thanaweya Amma), which qualifies them for admission to higher education institutions. Admission is primarily determined by performance on the national secondary school leaving examination, and most students who achieve the required scores continue to university education (Krafft, 2018). Students who complete the three-year vocational track and pass the final examination receive a Secondary School Technical Diploma in a chosen specialization such as commerce, industry, or agriculture. Likewise, students who complete the five-year vocational track and pass the final examination are awarded a Diploma of Advanced Technical Studies in commerce, industry, or agriculture (Kemper and Renold, 2024; Krafft, 2018). Although graduates from both the general and vocational tracks may apply to post-secondary programs in related fields, most vocational graduates enter the labor market directly. Only around 9 percent continue to higher education, most often through two-year post-secondary technical institutes (Krafft, 2018).

2.2 Gender role attitudes in Egypt

Our study examines how attending general versus vocational secondary schools affects attitudes towards women’s work and leadership roles in Egypt. To contextualise this

analysis, we first assess how Egyptian gender role attitudes compare internationally. According to the World Values Survey (2017 to 2022), Egypt ranks among the most conservative countries globally in attitudes towards women’s workforce participation and leadership ⁷. Among Egyptians surveyed, 89% agreed or strongly agreed that *when jobs are scarce, men should have more right to a job than women*, the highest level of agreement among 92 countries surveyed, including other nations in the Middle East and North Africa (MENA) region. Similarly, 82% of Egyptian respondents agreed that *men make better political leaders than women*, again the highest globally. Additionally, 74% agreed that *men make better business executives than women*, second only to Pakistan at 76%. These figures underscore the deeply entrenched patriarchal norms within Egyptian society.

The survey also reveals substantial gender differences in attitudes within Egypt. While 86% of men agreed that men make better political leaders, only 74% of women held this view. Similarly, 83% of men agreed that men make better executives, compared to 63% of women. This gender gap highlights that Egyptian women hold moderately more egalitarian views than men, though conservative attitudes remain prevalent among both genders. The prevalence of traditional gender norms makes Egypt a particularly important context for investigating how institutional factors, especially education, can influence these beliefs. By comparing general and vocational secondary education tracks, we examine whether different educational pathways challenge or reinforce conservative gender attitudes.

3 Data and Methodology

3.1 Data

We use data from the Egypt Labour Market Panel Survey (ELMPS), a nationally representative longitudinal household survey conducted by the Economic Research Forum (ERF) in partnership with Egypt’s Central Agency for Public Mobilization and Statistics (CAP-

⁷The data on this is accessible from the World Values Survey’s online analysis tool: [Click here](#)

MAS). The survey tracks individuals and households over time, covering a broad range of socioeconomic and demographic characteristics including educational attainment and gender role attitudes. Our analysis uses data from the 2018 and 2012 rounds of the survey. We take preparatory school test scores from 2012 round, the only round that collected this information, and gender role attitudes from 2018⁸, as the 2012 round does not provide information on gender role attitudes. We assess attitudes towards women’s participation in labour force and holding leadership positions through an individuals’s agreement with the two survey statements: (i) A woman’s place is not only in the household, but she should also be allowed to work’ and (ii) Women should continue to obtain leadership positions in society. Respondents rate their agreement with each statement on a five-point Likert scale ranging from *Strongly Disagree* to *Strongly Agree*. For each statement we construct a binary indicator of agreement, equal to 1 if the respondent agrees or strongly agrees with the statement and 0 otherwise.

We match the preparatory exam scores of an individual from 2012 with their GRAs in 2018 using unique individual identifiers. Following this matching procedure, our final sample consists of 3590 individuals aged between 18 to 49 years⁹, who achieved passing scores between 50 and 100 on the preparatory school exam, and subsequently completed secondary education. Applying the optimal bandwidth selector of [Calonico et al. \(2014\)](#) yields an average bandwidth of 7 points below and 14 points above the cutoff, giving an estimation sample of 1,813 individuals. The sample is nearly evenly split by gender, with 49.8% women and 50.2% men. By school type, 45.0% graduated general secondary school and 55.0% graduated vocational secondary school (Appendix Table [A1](#)).

⁸Gender role attitude questions are asked of all individuals aged 15 years and older.

⁹The upper age bound of 49 years is determined by the 1981 reform that introduced the end-of-preparatory school exam, as this represents the oldest cohort affected by the policy at the time of the 2018 survey ([Kemper and Renold, 2024](#))

3.2 Empirical strategy

We exploit a discontinuity in the assignment of students to general versus vocational secondary school based on their preparatory school exam score. In Egypt, students who score 70 or above on the end-of-preparatory exam are eligible to attend general secondary school, while those who score below are assigned to vocational secondary school. This institutional rule generates quasi-random variation in the type of secondary schooling completed around the cutoff, which we use to identify the causal effect of completing general versus vocational secondary school on gender role attitudes. However the compliance with this rule is imperfect, therefore, we use a fuzzy regression discontinuity design (FRDD). As opposed to a sharp RDD, which identifies the average treatment effect at the cutoff, the FRDD identifies the local average treatment effect (LATE) for compliers in the neighbourhood around the cutoff ([Shen and Zhang, 2016](#)). In the first stage, we estimate the effect of eligibility for general secondary education (scoring ≥ 70 on the preparatory exam) on the probability of completing general rather than vocational secondary education (equation 2). In the second stage, we use predicted values from the first stage to estimate the effect of completing general secondary education on gender role attitudes (equation 1).

Since both the treatment and outcome variables in our analysis are binary, we estimate a recursive bivariate probit model. When the treatment is endogenous and both variables are binary, nonlinear binary response models outperform the linear probability model (LPM) in terms of efficiency and bias ([Wooldridge, 2010](#); [Horrace and Oaxaca, 2006](#)), and provide the additional advantage of predicted probabilities bounded within zero and one. We estimate the following system of equations:

$$Y_i^* = \beta_0 + \beta_1 D_i + \beta_2 f(X_i) + \varepsilon_i, \quad Y_i = 1 \text{ if } [Y_i^* > 0] \quad (1)$$

$$D_i^* = \delta_0 + \delta_1 T_i + \delta_2 g(X_i) + v_i, \quad D_i = 1 \text{ if } [D_i^* > 0] \quad (2)$$

with

$$\begin{bmatrix} \varepsilon_i \\ v_i \end{bmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \right)$$

In equation 1, Y_i^* is a latent variable capturing the propensity of individual i 's agreement with each of the two statements: (i) *A woman's place is not only in the household, but she should also be allowed to work* and (ii) *Women should continue to obtain leadership positions in society*. We estimate separate regressions for each statement. D_i is an indicator equal to one if individual i completed general secondary education and zero if they completed vocational secondary education. The coefficient β_1 is our main parameter of interest, capturing the effect of completing general versus vocational secondary education on each attitude outcome.

Equation 2 is the first stage of the FRDD, where D_i^* is a latent variable capturing the probability of completing general secondary education, instrumented by T_i . T_i is an indicator equal to one if individual i is eligible for general secondary education, that is, if their preparatory exam score X_i is at or above the cutoff of 70, and zero otherwise. $f(X_i)$ and $g(X_i)$ approximate the functional form of the running variable in equations 1 and 2, respectively. For robustness, we allow the slope of the running variable to differ on each side of the cutoff. We assume the same functional form and bandwidth in both stages (Van der Klaauw, 2008), and cluster standard errors at the preparatory score level to account for the discrete nature of the running variable (Lee and Card, 2008).

3.2.1 Validity of the FRDD

The validity of the FRDD rests on two standard assumptions (Hahn et al., 2001; Ebeid and Oguzoglu, 2023). The first assumption requires a discontinuous jump in the probability of treatment at the cutoff. To assess this assumption, Figure 2 presents the first-stage discontinuity plot, showing the relationship between preparatory exam scores and the probability of completing general secondary education. The share of students completing

general secondary education rises sharply at the cutoff. Just below 70, only 16% of students complete the general track, compared to 37% at the cutoff. Beyond the threshold, the probability increases only gradually, indicating that not all eligible students enrol in or complete general secondary education. The presence of non-compliance on both sides of the cutoff confirms that a fuzzy rather than sharp design is appropriate.

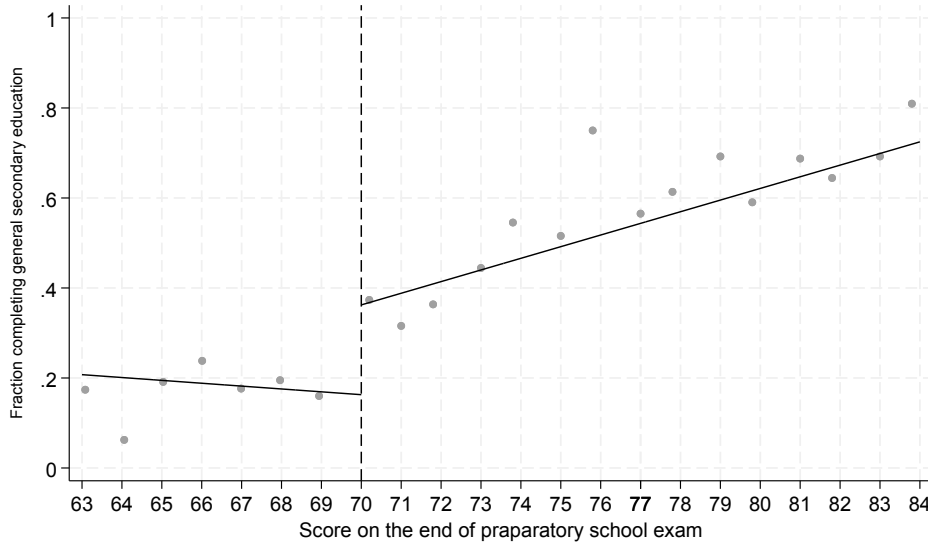


Figure 2: Preparatory exam score and the probability of completing general secondary education

Note: The figure plots the probability of completing general secondary education against the preparatory exam score within the optimal bandwidth. Each point represents the share of students completing general secondary education at a given score value. The dashed vertical line marks the eligibility cutoff at 70. Survey weights applied.

The second assumption requires that individuals cannot precisely sort around the cutoff ([Lee and Lemieux, 2010](#)), so that the assignment of students to either side of the threshold is as good as random in a neighbourhood around 70. If students are able to precisely manipulate their exam scores, those just above and just below may differ systematically, which would invalidate the design. We assess this assumption through three complementary tests.

We begin by examining the distribution of preparatory exam scores. Figure 3 shows a visible spike at 70, however similarly sized spikes appear at other five-point values throughout the distribution. As noted by [Barreca et al. \(2016\)](#) and [Cattaneo et al. \(2023\)](#), mass points of this kind arise when many observations share the same value of the running

variable, which in our setting likely reflects a common pattern among Egyptian teachers of grading in five-point steps, or recall bias among survey respondents. Notably, spikes of similar or larger magnitude appear at the surrounding five-point mass points, which suggests the spike at 70 is part of a broader distributional feature rather than evidence of manipulation at the threshold.

To formally test for manipulation, we apply the manipulation test using the local polynomial density estimators proposed by [Cattaneo et al. \(2020\)](#) to the full score distribution and get a T value of 5.7948 which is statistically significant at the 5% level. However, because the running variable is discrete with pronounced mass points, a significant density jump can arise from the mass-point structure of the score distribution rather than from selective sorting at the threshold ([Cattaneo et al., 2023](#)). To examine this, we conduct two additional tests. First, we re-run the manipulation test at 70 using only the mass points of the running variable, this isolates the variation at five-point intervals. This test gives a T value of 0.6616 which is not statistically significant at the 5% level, suggesting that the discontinuity at 70 is driven by the mass-point structure of the data rather than by manipulation. Second, following [Kemper and Renold \(2024\)](#) and [Dube et al. \(2019\)](#) we estimate the manipulation test at all five-point cutoffs from 55 to 95. As shown in [Figure A2](#), the T values exceed the 5% critical value at all placebo cutoffs, further confirming that the significant discontinuity at 70 is not unique to the policy threshold. Third, following [Schudde and Scott-Clayton \(2016\)](#), we re-estimate the manipulation test after excluding observations within one score point on either side of the cutoff. This test also does not detect a statistically significant discontinuity. Taken together, these results suggest that the significant estimate in the full-sample test reflects the discrete mass-point structure of exam scores rather than manipulation at the policy threshold.

As a final and more direct test of the no-sorting assumption, we examine whether predetermined covariates vary smoothly around the cutoff ([Lee and Lemieux, 2010](#)). If individuals were sorting around the threshold, we would expect those just above and just below 70 to differ systematically on predetermined characteristics. To assess this, we estimate reduced-

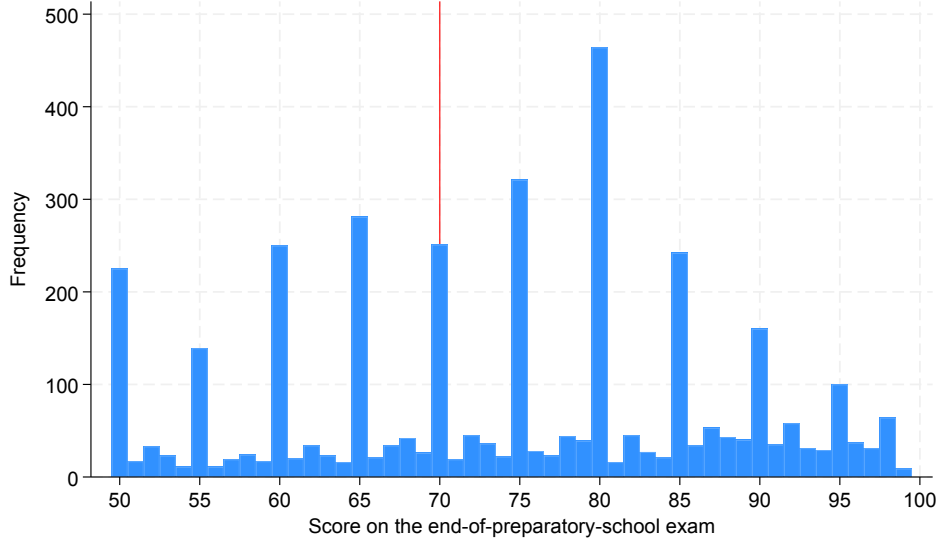


Figure 3: Frequency distribution of preparatory exam scores

Note: The histogram displays the frequency distribution of preparatory exam scores. The red dashed line marks the policy cutoff at 70.

form specifications within the optimal bandwidth using each predetermined covariate as the outcome. The covariates include gender, primary school exam score, age at completion of preparatory school, rural or urban birthplace, whether the individual’s mother was in paid employment when the individual was 15 years old, and parental education levels. Summary statistics are reported in Appendix Table A2. As shown in Table 1, none of the covariates exhibit a statistically significant discontinuity at the cutoff. This finding provides strong evidence against sorting and supports the comparability of individuals on either side of the threshold. Taken together, the results of all three tests lead us to conclude that manipulation of the running variable around the cutoff is unlikely as is also suggested by [Kemper and Renold \(2024\)](#).

Table 1: Continuity of covariates test

	LATE							
	(1) Female	(2) Age at prep	(3) Primary score	(4) Birth place	(5) Mother emp	(6) Mother edu	(7) Father edu	(8) Age
Scored ≥ 70	-0.022 (0.019)	0.120 (0.189)	-1.330 (1.548)	0.021 (0.044)	-0.011 (0.018)	-0.032 (0.029)	-0.033 (0.050)	-0.618 (0.874)
Observations	1844	1552	1186	1829	1840	1843	1844	1844
Control mean	0.432	14.08	73.70	0.557	0.111	0.134	0.233	32.59

The table reports reduced-form estimates testing for balance in predetermined covariates at the cutoff score of 70. Each column presents results from a separate regression where the reported coefficient corresponds to the effect of scoring at or above 70 on the preparatory exam. Covariates are: (1) gender, (2) age at completion of preparatory school, (3) primary school exam score, (4) rural birthplace, (5) mother in paid employment at age 15, (6) mother completed at least intermediate education, (7) father completed at least intermediate education, and (8) age in 2018. All estimates use survey weights. Standard errors in parentheses are heteroskedasticity-robust and clustered at the test score level. The control mean is the average of the outcome among individuals scoring below 70 within the optimal bandwidth.

4 Results

4.1 Effect of secondary school track on gender role attitudes

Table 2 reports the main results from the recursive bivariate probit model estimated within the optimal bandwidth. The estimates represent the local average treatment effect (LATE) of completing general rather than vocational secondary education on our two gender role attitude outcomes. Columns 1 and 2 report baseline estimates for attitudes toward women’s labour force participation and leadership positions respectively. Columns 3 and 4 report these estimates when allowing the regression function to vary on either side of the cutoff. Column 5 reports the first-stage estimates from a linear two-stage least squares (2SLS) model, while Columns 6 and 7 report the corresponding second-stage estimates for work and leadership outcomes respectively.

Our results suggest that completing general rather than vocational secondary education has a positive and statistically significant effect on both attitudes. Individuals who completed general secondary education are approximately 16 percentage points more likely to agree that women should be allowed to work and roughly 24 percentage points more likely to support that women should be in leadership positions. This represents a 21%

Table 2: Effect of completing general secondary education on attitudes towards women's labour force participation and leadership

	Baseline		Interaction		IV 2SLS		
	(1) Work	(2) Leadership	(3) Work	(4) Leadership	(5) First stage	(6) Work	(7) Leadership
Gen. secondary	0.160** (0.080)	0.244*** (0.049)	0.161** (0.081)	0.243*** (0.051)			
General secondary						0.247* (0.147)	0.330*** (0.119)
Scored ≥ 70					0.273*** (0.072)		
Control mean	0.76	0.68	0.76	0.68	0.19	0.76	0.68
F-Statistic					69.42		
Observations	1813	1813	1813	1813	1813	1813	1813

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

The table presents estimates from equations 1 and 2, capturing the local average treatment effect of completing general versus vocational secondary education on gender role attitudes, estimated using a recursive bivariate probit model within the optimal bandwidth. Column 1 reports the baseline estimate for attitudes toward women's labour force participation and Column 2 reports the baseline estimate for attitudes toward women in leadership positions. Columns 3 and 4 report estimates from specifications that allow the slope of the running variable to differ on either side of the cutoff by interacting the eligibility indicator with the running variable. Column 5 reports the first-stage estimates from a linear 2SLS model, while Columns 6 and 7 report the corresponding second-stage estimates for work and leadership outcomes respectively. The F-statistic refers to the Kleibergen-Paap F-statistic for instrument strength obtained from the 2SLS model. All estimates use survey weights, and standard errors in parentheses are heteroskedasticity-robust and clustered at the test score level. The control mean is the average of the respective outcome among respondents scoring below the cutoff within the estimation bandwidth.

$(\frac{0.16 \times 100}{0.77})$ increase relative to the control mean for supportive attitude towards work and 35% $(\frac{0.24 \times 100}{0.68})$ increase for supportive attitude towards leadership. The estimates remain similar when allowing the regression function to vary on either side of the cutoff. The RD plots in Figure A3 further support these findings, showing a visible positive jump in both outcomes at the cutoff.

As standard tests for instrument strength are not available for non-linear IV models, we estimate a linear two-stage least squares (2SLS) model alongside the recursive bivariate probit to assess instrument relevance. The first-stage estimate in Column 5 shows that scoring at or above 70 on the preparatory exam increases the probability of completing general secondary education by 27 percentage points. Given that standard errors are clustered at the test score level, we report the Kleibergen-Paap F-statistic as the measure of instrument strength. Columns 6 and 7 report the second-stage estimates for work and leadership outcomes respectively. Consistent with the bivariate probit results the estimates remain positive and statistically significant.

4.2 Potential Mechanisms

4.2.1 During secondary school mechanisms

We first consider mechanisms that may operate while students are enrolled in secondary school.

Peer composition: Adolescence marks a shift in social influence. As young people begin to detach from their parents, peers such as friends and classmates become a primary reference for attitudes and behaviours (Bussey and Bandura, 1999). The peer environment encountered during secondary school may therefore shape gender-role attitudes in lasting ways. Since eligibility for general secondary education depends on achieving a higher score and lower-scoring students are sorted into vocational tracks. These students may systematically differ in their socioeconomic backgrounds such as parental education that

is highly correlated with educational achievements and at the same time are known to predict GRAs (Du et al., 2021; Kulik, 2002). Therefore, the peer environment in terms of their gender role attitudes may differ significantly between the two tracks.

While we do not directly observe the attitudes of respondents’ classmates, the likely attitudinal climate of each track can be inferred from predetermined socioeconomic characteristics that are known predictors of GRAs. As shown in Panel A of Appendix Table A2, vocational secondary graduates are, on average, more likely to have grown up in rural areas, have parents with lower educational attainment, and were less likely to have had a working mother at age 15 — all characteristics associated with more traditional GRAs (Du et al., 2021; Kulik, 2002). The peer environment in vocational tracks may therefore be one where traditional GRAs are more prevalent. Exposure to these different attitudinal peer environments during a formative developmental period may therefore partly account for the gap in GRAs observed between the two groups.

Curriculum: General secondary education includes courses such as history, psychology, and philosophy (Elbadawy, 2015), which are likely to encourage discussions on societal issues, including gender equality. Research indicates that classroom discussions about gender equality foster more egalitarian attitudes and behaviours (Dhar et al., 2022). This curriculum emphasis on social sciences may therefore contribute to the more progressive gender attitudes observed among general secondary graduates. In contrast, vocational secondary education prioritises professional training in commerce, industry, and agriculture, offering fewer opportunities for such discussions.

4.2.2 Post-secondary school mechanisms

We next consider downstream mechanisms that unfold after graduation.

Post-secondary education: In Egypt, vocational secondary schooling is largely a terminal qualification that channels graduates directly into the labour market, whereas the general track is designed as a pathway to tertiary education (Krafft, 2018; Assaad, 2010).

We therefore hypothesise that part of the attitudinal gap between tracks operates through differential access to post-secondary schooling. Prior research links higher education to more progressive gender-role attitudes (Du et al., 2021; Rivera-Garrido, 2022; Zhang and Zhu, 2024), so greater post-secondary attainment among general secondary graduates may contribute to their more egalitarian views on women’s work and leadership.

Consistent with this hypothesis, post-secondary attainment differs sharply across tracks in our sample. Approximately 80% of general secondary graduates completed post-secondary education, compared with only 6% of vocational graduates (Appendix Table A3). Applying the same fuzzy regression discontinuity design as in the main analysis, we find that completing general rather than vocational secondary education increases the probability of post-secondary completion by 68 percentage points (Table 3, column 1). This large margin suggests that post-secondary schooling is a salient downstream pathway associated with the more progressive attitudes observed among general secondary graduates.

Workplace gender composition: After graduation, track placement may also shape attitudes through the gender composition of graduates’ workplaces. Appendix Table A4 documents how segregated this environment is already at school. Vocational enrolment is highly gendered by branch: about 68% of commercial students are female, whereas industrial and agricultural branches are predominantly male (66% and 74% male, respectively). General secondary is less segregated because most students enrol in arts or science (about 63% and 22% of general enrolment, respectively), where branch-level splits are close to even (about 53% female in each). We hypothesise that gender segregation across secondary branches carries over into workplace segregation. Among men in our sample, workplace composition is consistent with this pattern: 70% of male general secondary graduates report at least one female colleague in their primary job, compared with only 34% of male vocational graduates (Appendix Table A3). This differential exposure may contribute to the attitudinal gap among men between tracks, since working alongside women is associated with more egalitarian gender-role attitudes (Meuleman et al., 2017;

Yadav and Saharan, 2025).

To test whether track placement causally shifts this margin, we estimate whether completing general rather than vocational secondary increases the probability that men report female co-workers in their primary job, using the same fuzzy regression discontinuity design as in the main analysis. Column 2 of Table 3 shows that completing general secondary increases the probability of having female co-workers in the primary job by 48 percentage points, suggesting that crossing the cutoff and completing general secondary substantially raises men’s exposure to female co-workers at work.

Table 3: Effect of completing general secondary education on socio-economic outcomes

	LATE	
	(1) Higher education	(2) Female coworkers
Gen. secondary	0.704*** (0.066)	0.400*** (0.128)
Control mean	0.26	0.37
Observations	1826	749

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

The table reports local average treatment effects of completing general (rather than vocational) secondary education on post-secondary completion and on having female co-workers in the respondent’s primary job. Estimates use the same fuzzy regression discontinuity sample and bandwidth as the main analysis (recursive bivariate probit). We treat these as outcomes on margins that may lie on pathways linking track choice to attitudes, not as formal mediators of the attitude LATE. Estimates use survey weights. Standard errors are clustered at the preparatory exam score. The control mean is the weighted mean of the outcome among respondents below the cutoff within the bandwidth.

4.2.3 Post-secondary attainment and the main effects

We next ask whether the main attitudinal effects operate only through post-secondary attainment. Table 4 re-estimates the local average treatment effect separately for respondents who did and did not complete post-secondary education.

The effect of completing general rather than vocational secondary education is positive in both subsamples for attitudes toward women’s work and leadership. For work, the estimate

is 13 percentage points among those who did not complete post-secondary education but is not statistically significant, whereas it is 66 percentage points among post-secondary graduates and is statistically significant at the 1% level. The effect is significantly larger in the post-secondary subsample. For leadership, the effect is positive and statistically significant in both subsamples, at 31 percentage points without post-secondary education and 63 percentage points with post-secondary education, and is again significantly larger among post-secondary graduates. These results suggest that post-secondary education is an important pathway, but general secondary education raises support for women’s leadership even among graduates who do not continue to post-secondary education.

Table 4: Effect of completing general secondary education on gender-role attitudes, by post-secondary completion

	Work		Leadership	
	(1)	(2)	(3)	(4)
	No postsecondary	Postsecondary	No postsecondary	Postsecondary
Gen. secondary	0.126 (0.154)	0.656*** (0.062)	0.306*** (0.068)	0.625*** (0.050)
Control mean	0.74	0.80	0.67	0.70
Observations	1092	721	1092	721
p-value (equality test)	0.001	.	0.001	.

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

The table reports local average treatment effects of completing general (rather than vocational) secondary education on gender-role attitudes in the same FRDD sample (recursive bivariate probit; optimal bandwidth), estimated separately for respondents without and with post-secondary education. Columns (1) and (2) refer to attitudes toward women’s work; columns (3) and (4) refer to attitudes toward women’s leadership. Survey weights; standard errors clustered at preparatory score. The p-value for the equality test is from a Z-test of whether the treatment effect differs between the no post-secondary and post-secondary subsamples within each outcome. A reported value of 0.001 indicates $p < 0.001$.

4.3 Gender and age specific estimates

4.3.1 Heterogeneity by gender

Research suggests that education may have a stronger influence in shaping gender attitudes of males as compared to females (Yüçetas and Carol, 2024). This is likely because women tend to hold more egalitarian views on gender roles at baseline, leaving less room for

attitude shifts. Whereas, men typically begin with less egalitarian baseline attitudes and education can serve as a catalyst for shifting these attitudes towards greater gender equality (Davis and Greenstein, 2009; Röder and Mühlau, 2014; Yüçetas and Carol, 2024). Furthermore, empirical evidence indicates that male adolescents are more receptive to peer influence (Widman et al., 2016) and experience greater peer pressure to endorse peer norms than female adolescents (Halimi et al., 2021). Given these gender differences in baseline attitudes and social influences, we examine whether the effect of secondary school type on these attitude outcomes varies by gender¹⁰.

Table 5: Effect of completing general secondary education on attitudes towards women’s work and leadership, by gender

	Work		Leadership	
	(1) Female	(2) Male	(3) Female	(4) Male
Gen. secondary	0.090 (0.060)	0.218* (0.131)	0.079* (0.043)	0.396*** (0.084)
Control mean	0.88	0.67	0.85	0.56
Observations	903	910	903	910
p-value (equality test)	0.373		0.001	

Note: *** p<0.01, ** p<0.05, * p<0.10

The table reports the estimates of completing general (rather than vocational) secondary education on attitudes towards women’s work and women’s leadership roles by gender using a bivariate probit model within the optimal bandwidth. Columns 1 and 2 present the estimates for women, while Columns 3 and 4 present the corresponding results for men. All estimates use survey weights, and standard errors (in parentheses) are heteroskedasticity robust and clustered at the test score level. Control mean refers to the weighted mean of the outcome for respondents of each gender strictly below the cutoff within the estimation bandwidth. The Wald test reports the χ^2 statistic for equality of the treatment effect across gender, where Columns 1 and 3 are compared for Work and Columns 2 and 4 are compared for Leadership.

Table 5 presents the local average treatment effects estimated separately for women and men. The effects are considerably larger among male graduates. Men who completed general rather than vocational secondary schooling are 22 percentage points more likely to support that women should be allowed to work and 40 percentage points more likely to support women in leadership roles, and both estimates are statistically significant. Among women, the corresponding effects are 9 percentage points for work and 8 percentage

¹⁰Due to data limitations, our analysis is restricted to binary gender categories (male and female).

points for leadership. The estimate for work is not statistically significant, while the estimate for leadership is significant at the 10% level. The equality test indicates a statistically significant difference between men and women for leadership, but not for work. These patterns are consistent with a ceiling effect, whereby women’s already more egalitarian baseline attitudes leave less scope for further change, together with men’s greater susceptibility to peer influence during secondary school ([Widman et al., 2016](#); [Halimi et al., 2021](#)).

4.3.2 Heterogeneity by age

Research indicates that the effects of general versus vocational education on labour market outcomes differ between the short and long term ([Hanushek et al., 2017](#)). Building on this insight, we examine whether the impact of educational track on gender attitudes persists or evolves over time by comparing outcomes across age cohorts. Specifically, we test whether older graduates (above median age) exhibit different treatment effects than younger graduates (below median age).

Table 6 presents the local average treatment effects for younger and older cohorts, split at the median age. The effects are positive and statistically significant in both cohorts for attitudes toward women’s work and leadership. For work, the estimate is 16 percentage points in the younger cohort and 28 percentage points in the older cohort. For leadership, the corresponding estimates are 21 and 28 percentage points. In both cases, the coefficient is larger for the older cohort, which may reflect the cumulative influence of mechanisms discussed earlier, such as post-secondary education and exposure to more gender-diverse workplaces, that unfold over the life course. However, these patterns are suggestive only, because the equality tests do not reject equal effects across cohorts for either outcome.

Table 6: Effect of completing general secondary education on attitudes towards women’s work and leadership, by age cohort

	Work		Leadership	
	(1) Younger cohort	(2) Older cohort	(3) Younger cohort	(4) Older cohort
Gen. secondary	0.164*** (0.057)	0.278*** (0.100)	0.213** (0.092)	0.282** (0.127)
Control mean	0.75	0.75	0.68	0.67
Observations	956	673	956	673
p-value (equality test)	0.322		0.661	

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

The table reports local average treatment effects of completing general (rather than vocational) secondary education on attitudes toward women’s work and leadership, estimated separately for younger respondents (below median age) and older respondents (above median age). Estimates use a recursive bivariate probit model within the optimal bandwidth. All estimates use survey weights. Standard errors in parentheses are heteroskedasticity-robust and clustered at the preparatory exam score. The control mean is the weighted mean of the outcome for respondents in each age cohort who scored below the cutoff within the bandwidth. The p-value for the equality test compares the treatment effect between younger and older cohorts within each outcome.

4.4 Robustness checks

We assess the robustness of our findings through several specification checks. Firstly, we examine how the estimated treatment effects behave across a range of bandwidth choices around the MSE-optimal selector. We implement this sensitivity check by systematically expanding and contracting the optimal window. Each step to the right increases the bandwidth by one point on both sides of the cutoff, while each step to the left decreases it by one point. As shown in Figure 4, the estimated effects remain positive and statistically significant across a wide set of bandwidths.

Second, we test robustness to the inclusion of predetermined covariates as controls in the main specification. These covariates are gender, primary school exam score, age at completion of preparatory school, rural or urban birthplace, whether the individual’s mother was in paid employment when the individual was 15 years old, and parental education levels. Columns 1 and 2 of Table 7 show that attending general secondary education continues to produce more egalitarian views on women’s work and leadership

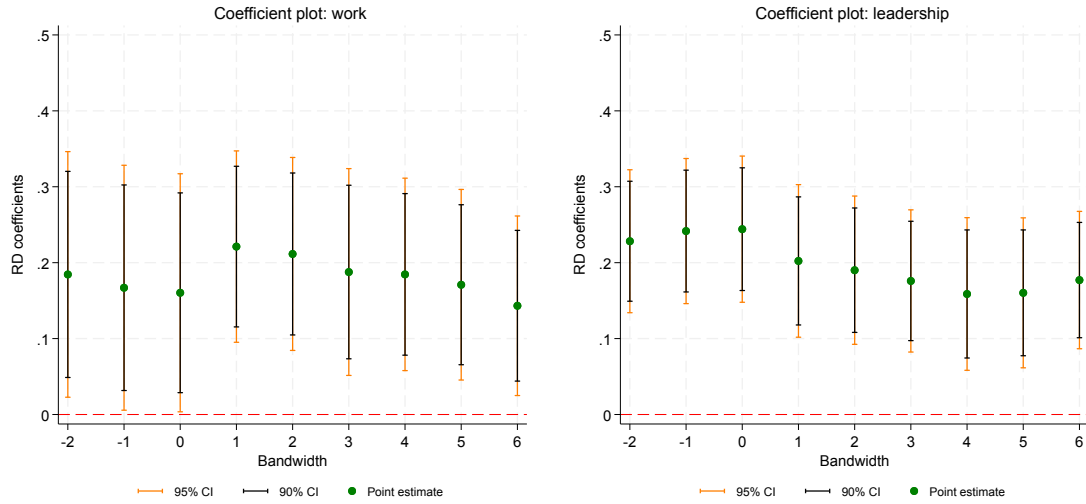


Figure 4: Effect of secondary school type on attitudes towards women’s work and leadership: changing bandwidths

The y-axis shows the LATE estimates of completing general secondary education on attitudes towards women’s work (left) and leadership roles (right), with 90% confidence intervals using bivariate probit model. The x-axis displays different bandwidths of the running variable, with zero marking the optimal bandwidth (asymmetric around the cutoff: -7 points below and +14 points above). Each step to the right (left) represents a one-point increase (decrease) in bandwidth width on both sides of the cutoff. For instance, 1 (–1) indicates adding (subtracting) a one-point increment to both sides of the optimal bandwidth around the cutoff. Standard errors are corrected for heteroskedasticity and allowing for clustering at test score level. Data is weighted using survey weights.

even after controlling for these baseline characteristics.

Third, we check sensitivity to the bandwidth choice. The baseline specification uses an asymmetric MSE-optimal window, while columns 3 and 4 of Table 7 re-estimate the model using a symmetric bandwidth that extends seven points on either side of the cutoff. The estimated effects remain positive and statistically significant, at about 30 percentage points for attitudes toward women’s work and 20 percentage points for leadership. This pattern is consistent with the baseline results and with the bandwidth sensitivity checks in Figure 4.

Finally, we report donut regressions that omit observations one exam score point below and above the cutoff, so that identification uses respondents located slightly farther from the threshold. If there were no manipulation around 70, this restriction should not significantly change the estimates (Barreca et al., 2016). Columns 5 and 6 of Table 7 show that completing general secondary schooling continues to have a significant positive effect on

Table 7: Effect of completing general secondary education on attitudes towards women’s labour force participation and leadership

	Added controls		Symmetric bandwidth		Donut regression	
	(1) Work	(2) Leadership	(3) Work	(4) Leadership	(5) Work	(6) Leadership
Gen. secondary	0.397* (0.239)	0.454*** (0.055)	0.297*** (0.083)	0.201** (0.081)	0.349*** (0.108)	0.320*** (0.066)
Control mean	0.79	0.69	0.78	0.71	0.77	0.71
Observations	935	935	1122	1122	833	833

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

The table reports local average treatment effects of completing general (rather than vocational) secondary education on attitudes toward women’s work and leadership. Columns 1 and 2 add predetermined covariates as controls within the optimal bandwidth. Columns 3 and 4 use a symmetric MSE-optimal bandwidth. Columns 5 and 6 repeat the symmetric-bandwidth specification but exclude observations one preparatory exam score point below and above the cutoff (833 observations). All estimates use survey weights. Standard errors are heteroskedasticity-robust and clustered at the preparatory exam score. The control mean is the weighted mean of the outcome among respondents below the cutoff within the relevant sample and bandwidth.

both gender-role attitude outcomes. The point estimates remain sizeable, and the overall pattern is qualitatively similar to the main results rather than being driven by individuals with scores immediately adjacent to the cutoff.

5 Conclusion

Vocational education has received growing policy attention in developing countries, where it is often promoted as a route into stable employment and higher earnings (Bol et al., 2019; Brunetti and Corsini, 2019). Much of the existing evidence therefore focuses on labour market outcomes, which is also what these programmes are primarily designed to deliver (Hanushek et al., 2017; Kemper and Renold, 2024). However secondary education track may also shape wider social outcomes by placing students in different learning environments during school and on different pathways after graduation. In this paper, we focus on one such outcome, gender role attitudes. These attitudes matter in their own right and are closely linked to women’s labour force participation and access to leadership

roles ([Patel et al., 2021](#); [Webb and Temple, 2016](#)).

Using Egyptian panel data and a fuzzy regression discontinuity design around the preparatory exam cutoff, we estimate the causal effect of completing general rather than vocational secondary education on support for women’s work and leadership. We find that completing general secondary education leads to more egalitarian attitudes regarding women’s participation in workforce and leadership positions. To our knowledge, this provides among the first causal evidence on how adolescent tracking in education streams shapes gender role attitudes in a developing-country setting. Plausible explanations include differences in the socioeconomic background of peers and in curriculum content during secondary school. Vocational tracks draw students from more disadvantaged backgrounds whose characteristics are associated with more traditional gender-role views ([Du et al., 2021](#); [Kulik, 2002](#)), while general secondary offers greater exposure to humanities-oriented coursework that can open discussion of social norms ([Elbadawy, 2015](#); [Dhar et al., 2022](#)). After graduation, general secondary graduates are also more likely to enter post-secondary education and, among men, to work alongside female co-workers, both of which may reinforce more egalitarian attitudes over time ([McHale, 1994](#); [Meuleman et al., 2017](#); [Yadav and Saharan, 2025](#)).

These findings carry policy relevance. In contexts where large shares of students enter vocational tracks, track placement is not only a skills decision but also a social one. Because gender role attitudes help shape female labour force participation, expanding discussion of society, history, and norms within vocational curricula may foster more egalitarian attitudes ([Dhar et al., 2022](#); [Patel et al., 2023](#)). So would policies that widen credible pathways from vocational secondary into post-secondary education, thereby reducing the sharp divide between terminal vocational training and broader higher education ([Krafft, 2018](#); [Assaad, 2010](#)).

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A Appendix

Table A1: Secondary education track by eligibility and gender

Scored ≥ 70	Female			Male		
	General	Vocational	Total	General	Vocational	Total
Total sample						
No	63 (11.65)	478 (88.35)	541 (100)	107 (15.07)	603 (84.93)	710 (100)
Yes	815 (66.80)	405 (33.20)	1220 (100)	731 (65.33)	388 (34.67)	1119 (100)
Total	878 (49.86)	883 (50.14)	1761 (100)	838 (45.82)	991 (54.18)	1829 (100)
Estimation sample						
No	35 (18.52)	154 (81.48)	189 (100)	46 (18.55)	202 (81.45)	248 (100)
Yes	393 (54.81)	324 (45.19)	717 (100)	348 (51.79)	324 (48.21)	672 (100)
Total	428 (47.24)	478 (52.76)	906 (100)	394 (42.83)	526 (57.17)	920 (100)

Table A2: Summary of pre-determined covariates

Variables	General		Vocational	
	Mean (SD)	Obs	Mean (SD)	Obs
Panel A: Total Sample				
Female	0.512 (0.500)	1716	0.471 (0.499)	1874
Age at prep. school graduation	14.03 (1.18)	1550	14.07 (1.51)	1562
Primary school score	85.86 (9.55)	1181	74.44 (11.68)	1096
Born in rural area	0.441 (0.497)	1683	0.577 (0.494)	1867
Mother in paid employment*	0.291 (0.454)	1713	0.084 (0.278)	1868
Mother edu: intermediate or higher	0.466 (0.499)	1716	0.117 (0.322)	1873
Father edu: intermediate or higher	0.584 (0.493)	1716	0.212 (0.409)	1874
Age	30.41 (7.39)	1716	32.76 (8.39)	1874
Panel B: Sample in optimal bandwidth				
Female	0.521 (0.500)	822	0.476 (0.500)	1004
Age at prep. school graduation	14.07 (1.10)	729	14.10 (1.53)	823
Primary school score	81.74 (8.50)	541	76.39 (9.56)	631
Born in rural area	0.471 (0.499)	809	0.547 (0.498)	1002
Mother in paid employment*	0.238 (0.426)	820	0.091 (0.287)	1002
Mother edu: intermediate or higher	0.376 (0.485)	822	0.132 (0.338)	1003
Father edu: intermediate or higher	0.495 (0.500)	822	0.213 (0.410)	1004
Age	31.18 (7.53)	822	33.02 (8.50)	1004

Notes: The table presents the summary of pre determined covariates using data from the ELMPS 2012 and ELMPS 2018 for individuals aged 25-49. In panel A, we use the whole sample whereas, we only use the observations within the optimal bandwidth in panel B.

*: The variable on whether mother was in paid employment or not corresponds to when the respondent was 15 years old.

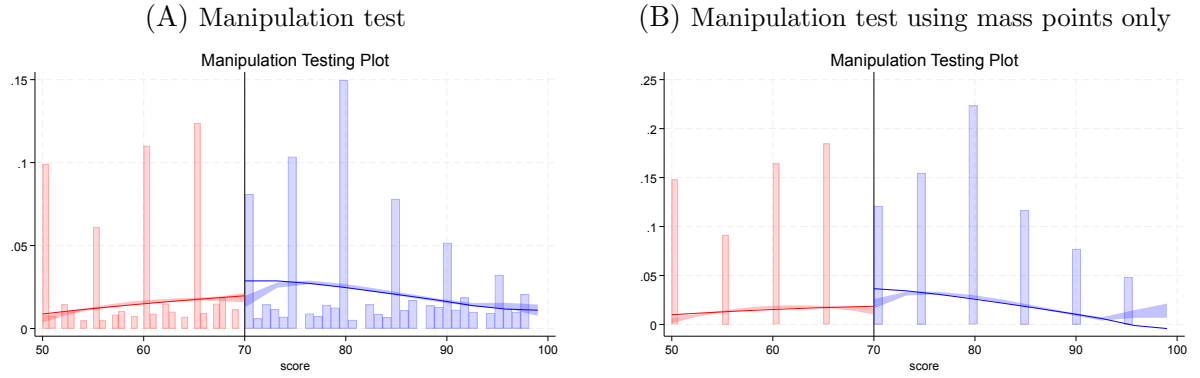


Figure A1: Manipulation Testing at the eligibility cutoff ($c = 70$)

Note: Panel A implements manipulation test using the local polynomial density estimators proposed by Cattaneo et al. (2020) to the full score distribution and get a T value of 5.7948 which is statistically significant at the 5% level. Panel B restricts the sample to five-point masspoints to isolate variation at heap values and get a T value of 0.6616 which is not statistically significant at the 5% level.

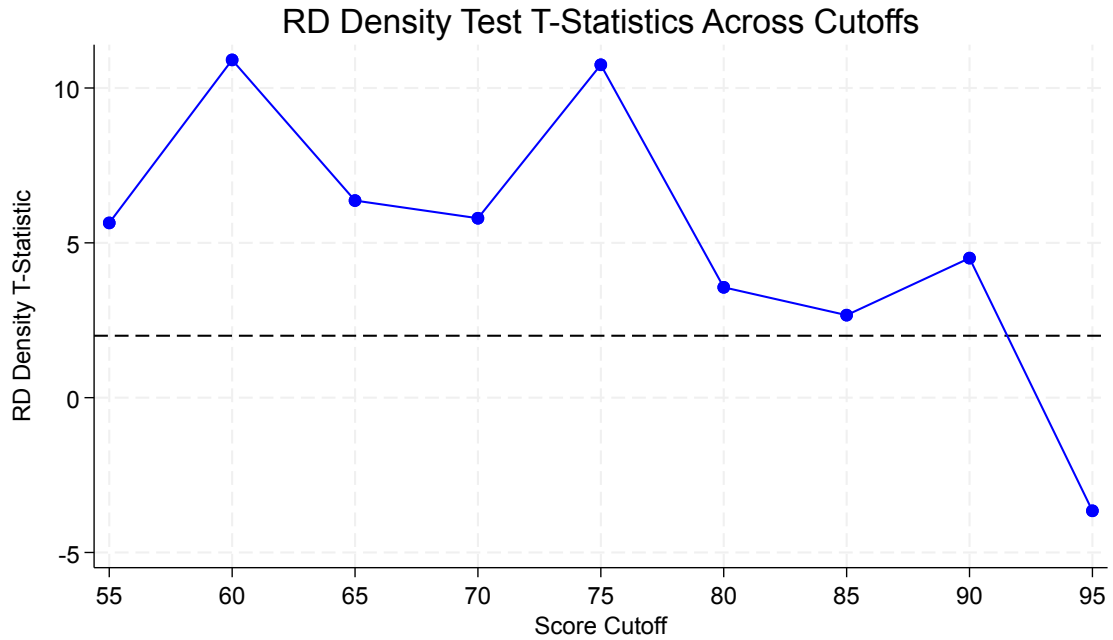


Figure A2: T-statistic based on manipulation test estimate by exam scores

Note: The y-axis shows the t-values associated with the manipulation test proposed by Cattaneo et al. (2020) at different placebo cutoffs. The x-axis displays the end-of-preparatory-school exam score. The black dashed horizontal line represents the critical value of the t-distribution at the 5% significance level.

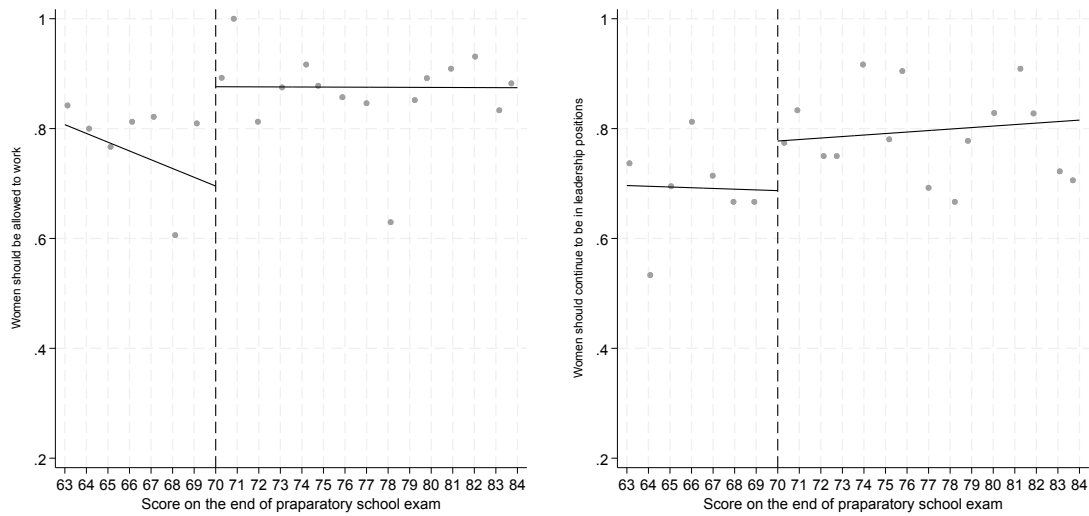


Figure A3: Effect of secondary school type on attitudes towards women's work and leadership: output plots

The figure plots the mean values of our outcome variables (y-axis) against the running variable (x-axis). Mean values are displayed within the optimal bandwidth and the fitted lines on either side of the cutoff show predicted outcomes from linear regressions

Table A3: Descriptive statistics of outcomes for the total and estimation sample

Variables	General		Vocational	
	Mean (SD)	Obs	Mean (SD)	Obs
Panel A: Total Sample				
Women should be allowed to work	0.881 (0.323)	1710	0.794 (0.404)	1860
Women should be in leadership positions	0.808 (0.394)	1710	0.701 (0.458)	1859
Post secondary education	0.798 (0.401)	1716	0.061 (0.239)	1874
Female coworkers (male subsample)	0.707 (0.455)	836	0.341 (0.474)	993
Panel B: Sample in optimal bandwidth				
Women should be allowed to work	0.867 (0.340)	819	0.803 (0.398)	994
Women should be in leadership positions	0.786 (0.410)	819	0.708 (0.455)	994
Post secondary education	0.792 (0.406)	822	0.073 (0.260)	1004
Female coworkers (male subsample)	0.653 (0.477)	415	0.378 (0.485)	535

Notes: The table presents the summary of outcome variables using data from the ELMPS 2012 and ELMPS 2018 for individuals aged 25-49.

Table A4: Gender composition in general and vocational secondary school branches

Gender	General			Vocational			Total
	Math's	Science	Arts	Commercial	Industrial	Agricultural	
Female	89	186	535	466	311	44	1,631
	(37.24)	(52.69)	(53.29)	(67.93)	(33.91)	(25.73)	(48.40)
Male	150	167	469	220	606	127	1,739
	(62.76)	(47.31)	(46.71)	(32.07)	(66.09)	(74.27)	(51.60)
Total	239	353	1,004	686	917	171	3,370
	(100)	(100)	(100)	(100)	(100)	(100)	(100)

Notes: The table reports counts by branch and gender. Percentages in parentheses are column percentages.